

# Zheyu Zhang

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## Education

### University of Waterloo

B.Sc. in Materials and Nanosciences, expected 2027; average: 92.64%

Sep. 2025 – Present

Waterloo, ON, Canada

### Beijing Jiaotong University

B.Eng. in Nanomaterials and Technology, expected 2027

Sep. 2023 – Present

Beijing, China

## Research Experience

### URA – Li<sup>+</sup> Ion Transport Modeling in LIPON

University of Waterloo, supervised by Prof. Conrad G. Tsetsis Feugmo

Jan. 2026 – Present

Waterloo, ON, Canada

- Implementing and analyzing a PNP-cDFT model for Li<sup>+</sup> ion transport in LIPON solid electrolytes.
- Analyzing temperature-dependent diffusion behavior and transport trends using literature-based parameters.
- Developing numerical simulations and data-analysis workflows in Python/JAX.

### URA – Fe-Alginate Microgels via Droplet Microfluidics

Beijing Jiaotong University, supervised by Prof. Yuandu Hu

Jun. 2024 – Apr. 2025

Beijing, China

- Investigated Fe-alginate microgel fabrication via droplet microfluidics coupled with external ionic crosslinking.
- Analyzed how droplet composition, collection-solution viscosity, and FeCl<sub>3</sub> concentration influence microgel morphology and deformation.
- Performed experimental analysis of microgel morphology, size distribution, and structure–property relationships.

## Publications

**Chen, J.**, Li, P., An, R., Cai, A., Wang, K., **Zhang, Z.**, Wang, T., Hu, Y. “Droplet Microfluidics-Assisted Fabrication of Fe-Alginate Microgels with Complex Morphology: Effect of the Composition of Droplets.” *Lab on a Chip*, 2025, 25(13), 3270–3286. DOI: 10.1039/D4LC01101E.

**Chen, J.**, Yu, R.-Y., Wang, K.-Q., **Zhang, Z.-Y.**, Ardekani, A., Hu, Y.-D. “Spherical Magnetic Fe-Alginate Microgels Fabricated by Droplet-Microfluidics Combining with an External Crosslinking Approach and the Study of Their pH-Dependent Fe<sup>3+</sup> Release Behaviors.” *Chinese Journal of Polymer Science*, 2025, 43, 289–302. DOI: 10.1007/s10118-025-3257-2.

## Technical Skills

**Programming & Modeling:** Python, JAX, NumPy, Matplotlib, Li<sup>+</sup> transport modeling

**Materials & Characterization:** droplet microfluidics, optical microscopy, SEM

**Research Tools:** LaTeX, Origin, MATLAB

**Languages:** Mandarin Chinese (Native), English (Fluent)